Teacher-Free Adaptive Inquiry Under World Change

Anonymous Authors

Abstract

Suite C previously showed that a hand-specified policy and then a teacher-trained probe head can re-engage after world change. This paper reports the next finite step: a linear inquiry policy selected by teacher-free cross-entropy-method reward search. The training objective uses downstream recovery, selective re-engagement, second-shift reopenability, probe cost, and anti-cheat controls; it does not consume hand-policy labels, teacher actions, or teacher probabilities. On held-out seeds the policy passes C1-C6 plus T1/N1, and a 64-seed bootstrap replication preserves the effect. A follow-up ablation replaces the privileged source-identity bit with an observable source estimate and routes the policy through a local tool-call adapter; both finite gates pass, while a malformed-tool control fails.

1 Introduction

The reviewer objection is direct: imitation of a hand-designed inquiry law is weaker than teacher-free adaptive inquiry. The result here removes direct teacher supervision while keeping the same minimum pass rule: final recovery is not sufficient unless inquiry is selective, fresh, cost-aware, reopenable, and robust to anti-cheat controls.

2 Suite C Task

Describe two world shifts, affected and unaffected buckets, probe cost, recovery, false calm, and second-shift reopenability.

3 Teacher-Free Training

The implemented policy is a linear probe head over the existing observable Suite C feature surface: perceived error, perceived surprise, jumps relative to a lagging baseline, effort, improvement, time since probe, recent probe rate, source identity, and bucket index. Cross-entropy-method search selects parameters using only downstream reward terms for recovery, selectivity, reopenability, probe cost, no-false-calm behavior, and control failure.

4 Gates and Controls

The accepted policy must pass C1-C6 plus T1 and N1. T1 audits that no teacher labels, teacher actions, or teacher probabilities enter the training loss. N1 requires stale-signal, wrong-signal, signal-suppression, and matched-random controls to fail in the intended ways.

5 Results

The first held-out run passes all gates: final affected MAE 0.095, recovery rate 1.000, first-shift selectivity 12.500, second-shift reopenability 11.312, and 22.0 probes. Matched random at the same budget reaches selectivity 0.833; stale-signal recovery is 0.000; wrong-signal selectivity is 0.000; and suppressed-signal final MAE is 0.464.

A wider 64-seed bootstrap replication preserves the conclusion. Final affected MAE is 0.095 with 95% CI [0.095, 0.096]; recovery rate is 1.000 [1.000, 1.000]; selectivity lift over matched random is 11.292 [10.976, 11.596]; and suite pass rate is 1.000 [1.000, 1.000].

6 Limitations

Finite simulator, no biological or consciousness claim, no production-agent claim, and no external API-agent result. The new source-estimate/tool-transfer run removes the privileged source bit inside this harness, but external open-model replication remains open.